INCH-POUND
MIL-DTL-1183/10B
w/AMENDMENT 1
20 August 2019
SUPERSEDING
MIL-DTL-1183/10B
20 December 2016

DETAIL SPECIFICATION SHEET

FITTINGS, PIPE, CAST BRONZE, SILVER-BRAZING; UNION, PIPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-1183.

REQUIREMENTS:

1. Female union dimensions. Dimensions for the female union components (see figures 1, 2, and 3) shall be as specified in table I. On figure 2, the dimensions P, Q, and R define the point of contact of the ground seat.

2. Union type. Female unions (socket both ends) shall be Type A (streamlined interior). The male tailpiece (see figure 4) shall be Type A (streamlined interior).

3. Male tailpiece dimensions. Dimensions for the threaded male tailpiece (see figure 4) shall be as specified in table I. Male tailpiece dimensions are the same as the dimensions for the female tailpiece (see figure 2), except for those dimensions identified in figure 4. NOTE: The size of the male pipe threads is one size larger than the nominal pipe size by which the threaded male tailpiece is identified.

4. Nut dimensions. Dimensions for the union nut (see figure 3) shall be as specified in table I.

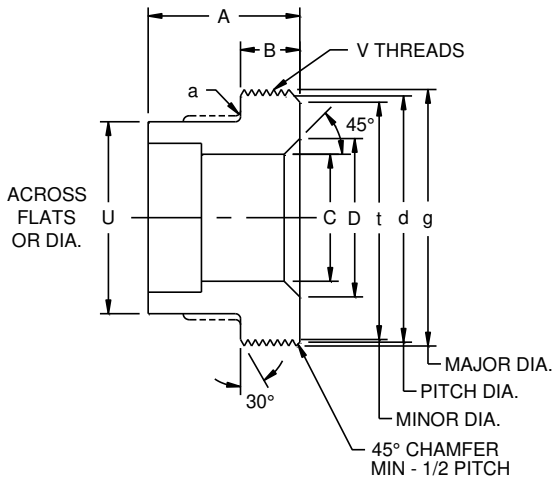


FIGURE 1. Female thread piece.

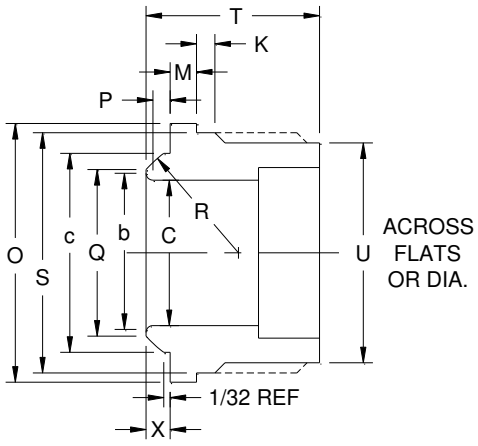


FIGURE 2. Female tailpiece.

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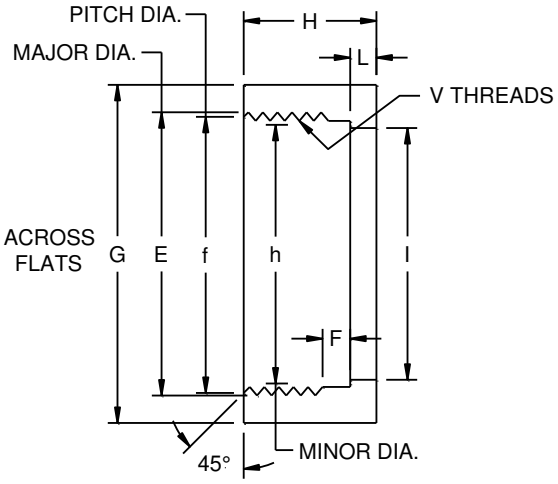


FIGURE 3. Nut.

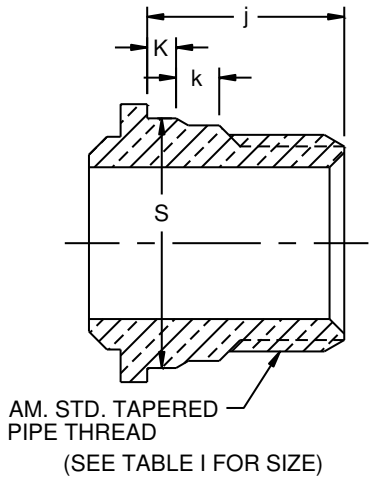


FIGURE 4. Male thread tailpiece.

TABLE I. Union components dimensions, inches.

Nominal pipe size	A ±0.02	B ±1/32	C Diameter		D Diameter	E	F ±0.016	G	H Min	I Diameter
¼	0.85	23/64	0.398	0.015	0.610	1.095	0.120	1 11/32	21/32	0.860
⅜	0.94	25/64	0.532	±0.015	0.755	1.281	0.125	1 35/64	45/64	1.020
½	0.99	25/64	0.697	±0.020	0.960	1.491	0.135	1 25/32	47/64	1.220
¾	1.18	½	0.907	±0.025	1.140	1.835	0.150	2 3/16	7/8	1.510
1	1.26	½	1.171	±0.025	1.445	2.165	0.170	2 35/64	59/64	1.820
1¼	1.43	35/64	1.502	±0.030	1.770	2.601	0.200	3 3/64	1 1/32	2.220
1½	1.52	35/64	1.742	±0.030	2.060	2.921	0.220	3 25/64	1 1/16	2.520
2	1.67	35/64	2.186	±0.030	2.545	3.521	0.250	4 3/64	1 5/32	3.090

TABLE I. Union components dimensions, inches – Continued.

Nominal pipe size	K ±0.016	L Min	M ±0.010	O Diameter	P	Q Diameter	R Radius	S Diameter	T ±0.020
¼	0.125	0.125	0.125	0.965	0.100	0.510	0.361	0.83	0.874
⅜	0.135	0.135	0.135	1.135	0.103	0.650	0.460	0.99	0.959
½	0.145	0.145	0.145	1.345	0.113	0.835	0.591	1.19	1.049
¾	0.160	0.160	0.160	1.650	0.115	1.030	0.728	1.47	1.154
1	0.175	0.175	0.180	1.975	0.135	1.315	0.930	1.78	1.299
1¼	0.205	0.205	0.210	2.395	0.130	1.650	1.167	2.18	1.469
1½	0.220	0.220	0.230	2.710	0.143	1.915	1.355	2.47	1.619
2	0.250	0.250	0.260	3.310	0.165	2.395	1.694	3.04	1.854

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TABLE I. Union components dimensions, inches – Continued.

Nominal pipe size	U	V	X	Pipe thread size	a Radius (ref)	b Diameter (ref)	c Diameter	d Pitch diameter	
								Min	Max
¼	0.75	16-UNS	0.134	⅜	0.094	0.441	0.618	1.0456	1.0506
⅜	0.90	14-UNS	0.139	½	0.094	0.576	0.768	1.2250	1.2304
½	1.09	14-UNS	0.159	¾	0.094	0.741	0.972	1.4348	1.4403
¾	1.35	11-UNS	0.154	1	0.125	0.951	1.174	1.7640	1.7702
1	1.63	11-UNS	0.184	1¼	0.125	1.216	1.494	2.0938	2.1001
1¼	2.00	10-UNS	0.174	1½	0.125	1.561	1.828	2.5233	2.5301
1½	2.27	10-UNS	0.199	2	0.125	1.801	2.116	2.8432	2.8501
2	2.80	10-UNS	0.224	2½	0.125	2.276	2.638	3.4430	3.4499

TABLE I. Union components dimensions, inches – Continued.

Nominal pipe size	f Pitch diameter		g Major diameter		h Minor diameter		j	k (ref)	t Minor diameter
	Min	Max	Min	Max	Min	Max			Max
¼	1.0520	1.0585	1.0846	1.0940	1.0278	1.0348	0.985	0.30	1.0133
⅜	1.2320	1.2390	1.2697	1.2800	1.2043	1.2122	1.155	0.30	1.1884
½	1.4420	1.4492	1.4764	1.4867	1.4111	1.4190	1.165	0.30	1.3984
¾	1.7720	1.7801	1.8172	1.8293	1.7326	1.7424	1.350	0.31	1.7175
1	2.1020	2.1102	2.1471	2.1592	2.0626	2.0724	1.505	0.33	2.0475
1¼	2.5321	2.5407	2.5821	2.5950	2.4887	2.4992	1.535	0.33	2.4723
1½	2.8520	2.8609	2.9021	2.9150	2.8087	2.8192	1.610	0.39	2.7923
2	3.4520	3.4610	3.5020	3.5149	3.4087	3.4192	2.020	0.39	3.3923

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5. Static O-ring seal and retainer ring. When specified (see requirement 8 herein), unions shall be provided with an O-ring and a retainer ring which, when installed, is outboard of the ground joint (see figure 5). The O-ring and retainer do not affect the primary contact of the ground seat and form an effective secondary seal. NOTE: For Navy applications, unions shall not be used without the O-ring and retainer ring.

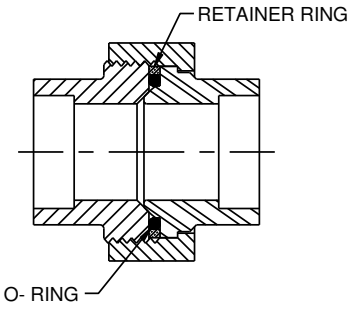


FIGURE 5. O-ring and retainer ring placement.

6. Retainer ring dimensions and material. Retainer ring dimensions shall be as specified in table II. The material of the retainer ring shall be ASTM B584 (UNS no. C90300) or ASTM B61 (UNS no. C92200) bronze. For prohibited materials, see paragraph 3.2 of the base document.

7. O-ring seal configuration and material. O-ring configuration shall be in accordance with SAE-AS568 of size given in table III for the required union nominal pipe size. O-ring material shall be in accordance with SAE-AMS-7276, Class 1 rubber and SAE-AS3208 and SAE-AS3209 for part number and sizing. For steam service to 300 °F, O-ring material shall be ASTM D2000, 2AA 815 A13F17EA14Z (300 °F steam and phosphate ester base hydraulic fluid service) ethylene propylene rubber.

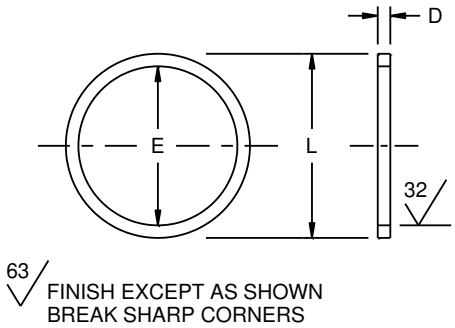


FIGURE 6. Retainer ring.

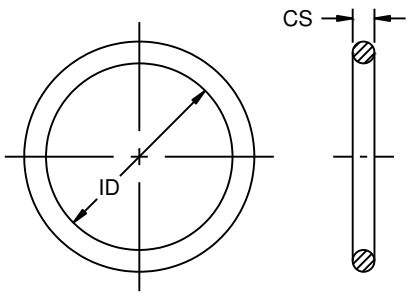


FIGURE 7. O-ring.

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TABLE II. Retainer ring dimensions, inches.

Nominal pipe size	D ±0.005	E ±0.005	L ±0.005
¼	0.050	0.813	1.008
⅜	0.050	0.938	1.184
½	0.050	1.188	1.394
¾	0.060	1.500	1.715
1	0.070	1.750	2.045
1¼	0.070	2.062	2.471
1½	0.070	2.375	2.791
2	0.090	3.000	3.391

TABLE III. O-ring configuration and size.

Nominal pipe size (inches)	O-ring configuration and size
¼	AS568-017
⅜	AS568-019
½	AS568-023
¾	AS568-125
1	AS568-129
1¼	AS568-134
1½	AS568-139
2	AS568-232

8. Specification sheet part number. The specification sheet part number is a definitive part number that is formed by combining the specification sheet code number (M1183/10 for MIL-F-1183/10) with a dash after it, the item and pipe size code number, the brazing ring groove and code letter, the O-ring and retainer ring code number, and an optional dash and lead-free fitting identifier. Unions covered by this specification sheet shall be designated in the following form (see 6.2.b of MIL-DTL-1183):

M1183/10 – XX X X - PW

Specification sheet code number

Item and pipe size code number (see requirement 9 herein)

Brazing ring groove and brazing ring code letter (see requirement 10 herein)

O-ring and retainer ring code number (see requirement 11 herein)

Lead-free fitting identifier (optional) (see requirement 6 herein)

9. Item and pipe size. The item (female union, male tailpiece, and union nut) and the nominal pipe sizes are identified by two digits (see table IV).

TABLE IV. Code number to item and size.

Female union (figures 1, 2, and 3)	Male tailpiece (figure 4)	Union nut (figure 4)	Nominal pipe size (inches)
10	30	50	¼
11	31	51	⅜
12	32	52	½
13	33	53	¾
14	34	54	1
15	35	55	1¼
16	36	56	1½
17	37	57	2

10. Brazing ring groove and brazing ring. No groove (face fed, male tailpiece, or nut), grooved without brazing ring, grooved with brazing ring inserted in groove, grooved with brazing rings separately packaged and attached to the union, or grooved with brazing rings separately packaged and shipped with but not attached to the union socket ends are identified by a single letter (see table V).

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TABLE V. Code letter to socket condition.

Code letter	Condition of socket
N	No groove (face fed)
G 1/	Grooved (no brazing ring)
R 1/	Grooved with brazing ring inserted
A 2/	Grooved with brazing ring separately packaged and attached
S 1/ 2/	Grooved with brazing ring separately packaged and shipped with

NOTES:

1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for a union with the code letter of G, R, or S.

2/ When the socket condition code letter is A or S, the brazing rings shall not be inserted in the grooves, but shall be packaged separately. For condition code letter A, the packaged brazing ring shall be either included within each individual union package or physically attached to the union. For condition code letter S, the packaged brazing ring shall be included within each individual bend package.

11. O-ring seal and retainer ring. Whether the item is provided without O-ring and retainer ring, with O-ring and retainer ring installed, with O-ring and retainer ring separately packaged and attached to the item, or with O-ring and retainer ring separately packaged and shipped with the item but not attached to item, it is identified by a digit (see table VI).

TABLE VI. Code number to O-ring and retainer ring requirement.

Code number 1/	O-ring and retainer ring requirement
1	No O-ring and retainer ring (for male tailpiece and union nut only)
2	Without O-ring and retainer ring (for female union only)
3	With O-ring (air, oil, or water service) and retainer ring installed 2/
4	With O-ring (steam service) and retainer ring installed 2/
5	With O-ring (air, oil, or water service) and retainer ring separately packaged and attached
6	With O-ring (steam service) and retainer ring separately packaged and attached
7	With O-ring (air, oil and water service) and retainer ring separately packaged and shipped with
8	With O-ring (steam service) and retainer ring separately packaged and shipped with

NOTES:

1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for a female union with a code number of 1, 2, 5, 6, 7, or 8 and for a male tailpiece or union nut with code number 2, 3, 4, 5, 6, 7, or 8.

2/ This requirement is not applicable to male tailpiece (see figure 4) or union nut (see figure 3).

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CONCLUDING MATERIAL

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Navy – SH
(Project 4730-2019-030)

Review activity:
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